

# Erdős Problem #521

EP-521

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

## ORIGINAL PROBLEM STATEMENT

Let  $(\epsilon_k)_{k \geq 0}$  be independently uniformly chosen at random from  $\{-1, 1\}$ . If  $R_n$  counts the number of real roots of  $f_n(z) = \sum_{0 \leq k \leq n} \epsilon_k z^k$  then is it true that, almost surely,  $\lim_{n \rightarrow \infty} \frac{R_n}{\log n} = \frac{2}{\pi}$ ?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>